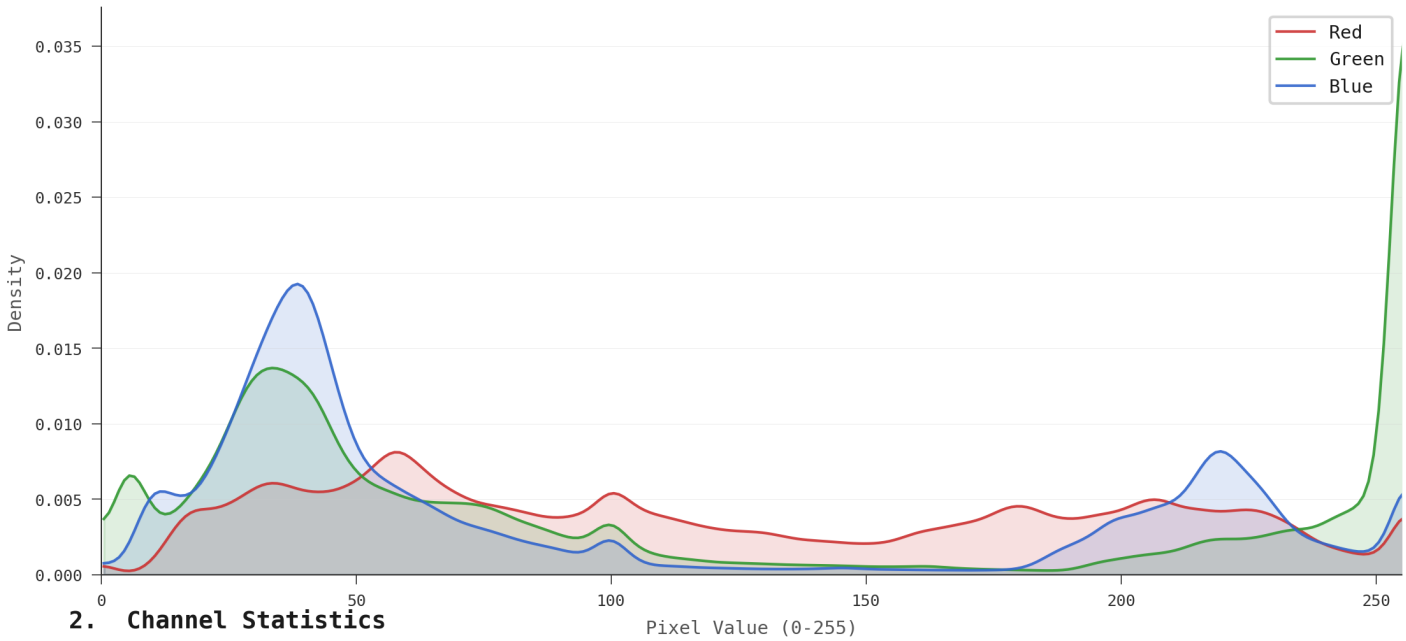
Reference Data Sheet: KODIM15

Subject: Face Paint Girl Portrait

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	119.2734	71.0346	5045.9174	-1.2877	0.2573	0	255
Green	105.7720	91.8071	8428.5387	-1.2306	0.6876	0	255
Blue	95.8575	82.3449	6780.6804	-1.1482	0.7660	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.8602	0.8562
G	0.8602	1.0000	0.9880
B	0.8562	0.9880	1.0000

Pairwise Pearson Correlation Coefficients

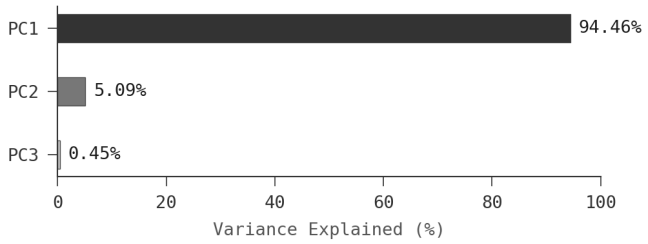
r(R-G)	=	**0.860191**
r(R-B)	=	**0.856245**
r(G-B)	=	**0.987986**
Avg |r|	=	**0.901474**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{5045.9302} & 5609.7318 & 5008.4767 \\
5609.7318 & \mathbf{8428.5601} & 7469.0397 \\
5008.4767 & 7469.0397 & \mathbf{6780.6977}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM15

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.4709	**-0.6573**	-0.5884	19133.4408
PC2	-0.8821	0.3431	0.3227	1031.5917
PC3	0.0103	-0.6710	0.7414	90.1555

6. Derived Metrics

Condition Number

212.23

(l1/l3) PC1/PC2: 18.55 | PC2/PC3: 11.44

Blue Channel Independence

2.3%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6573 | Green-Blue coupled

Classification: Strongly One-Dimensional (PC1 = 94.46%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.981212	0.988164	0.984688
Green	0.988726	0.992572	0.990649
Blue	0.986591	0.991229	0.988910

8. Average Local Variance (3x3)

Channel	Value
Red	124.1846
Green	130.6691
Blue	120.9060

9. Redundancy Profile

Classification: Strongly One-Dimensional | Condition Number: 212.23 | PC1: 94.46%

Correlation: G–B pair dominates at r = 0.987986 | R–G: 0.860191 | R–B: 0.856245

Blue Channel: 2.3% independent | 97.7% captured by PC1

Spatial: Avg autocorrelation: 0.988082 | Avg local variance: 125.2532

Eigenvector: Green-Blue coupled | G (-0.6573), B (-0.5884) co-load | R offset (-0.4709)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."